Lab Values Cheat Sheet

Adult reference intervals in US conventional and SI units, conversion factors, nursing cues and unit traps for NCLEX review

MindShelf

TRAP Reference intervals vary between laboratories
Interpret a result against the interval printed on **your lab's report**. The values below are one published set, for adults, for study.

Adult reference intervals

Test	US conventional	Factor	SI
Electrolytes			
Sodium (Na^+)	136–145 mEq/L	× 1	136–145 mmol/L
Potassium (K^+)	3.5–5.0 mEq/L	× 1	3.5–5.0 mmol/L
Chloride (Cl^-)	98–106 mEq/L	× 1	98–106 mmol/L
Bicarbonate (HCO_3^-), serum	23–28 mEq/L	× 1	23–28 mmol/L
Total CO_2, serum	23–30 mEq/L	× 1	23–30 mmol/L
Calcium, total	8.6–10.2 mg/dL	× 0.2495	2.15–2.54 mmol/L
Calcium, ionized	4.65–5.25 mg/dL	× 0.2495	1.16–1.31 mmol/L
Magnesium	1.6–2.6 mg/dL	× 0.4114	0.66–1.07 mmol/L
Phosphorus (as P)	3.0–4.5 mg/dL	× 0.3229	0.97–1.45 mmol/L
Glucose			
Glucose, fasting plasma	70–99 mg/dL	× 0.0555	3.9–5.5 mmol/L
Hemoglobin A1c	4.0–5.6 %	IFCC eq.	20–38 mmol/mol
Kidney			
BUN (urea nitrogen) → urea	8–20 mg/dL	× 0.357	2.9–7.1 mmol/L
Creatinine, female	0.50–1.10 mg/dL	× 88.4	44–97 µmol/L
Creatinine, male	0.70–1.30 mg/dL	× 88.4	62–115 µmol/L
Blood count			
Hemoglobin, female	12–16 g/dL	× 10	120–160 g/L
Hemoglobin, male	14–18 g/dL	× 10	140–180 g/L
Hematocrit, female	37–47 %	× 0.01	0.37–0.47 L/L
Hematocrit, male	42–50 %	× 0.01	0.42–0.50 L/L
WBC (leukocytes)	4,000–11,000 /µL	× 0.001	4.0–11.0 ×10⁹/L
Platelets	150,000–450,000 /µL	× 0.001	150–450 ×10⁹/L
Coagulation			
PT (prothrombin time)	11–13 s	same	11–13 s
INR, no anticoagulant	0.8–1.1	same	0.8–1.1
INR target on warfarin	2.0–3.0	same	2.0–3.0
aPTT	25–35 s	same	25–35 s
Arterial blood gas (room air)			
pH	7.38–7.44	same	7.38–7.44
$PaCO_2$	38–42 mm Hg	× 0.1333	5.1–5.6 kPa
PaO_2	75–100 mm Hg	× 0.1333	10.0–13.3 kPa
HCO_3^-	23–26 mEq/L	× 1	23–26 mmol/L
SaO_2	≥ 95 %	× 0.01	≥ 0.95 (fraction)
Liver panel			
Albumin	3.5–5.5 g/dL	× 10	35–55 g/L
Total protein	5.5–9.0 g/dL	× 10	55–90 g/L
Bilirubin, total	0.3–1.0 mg/dL	× 17.1	5–17 µmol/L
ALT and AST (each)	10–40 U/L	× 0.0167	0.17–0.67 µkat/L
Alkaline phosphatase	30–120 U/L	× 0.0167	0.50–2.00 µkat/L
Lipids (decision limits)			
Total cholesterol, desirable	< 200 mg/dL	× 0.0259	< 5.18 mmol/L
LDL cholesterol, optimal	< 100 mg/dL	× 0.0259	< 2.59 mmol/L
HDL, low (female)	< 50 mg/dL	× 0.0259	< 1.30 mmol/L
HDL, low (male)	< 40 mg/dL	× 0.0259	< 1.04 mmol/L
Triglycerides, fasting	< 150 mg/dL	× 0.0113	< 1.70 mmol/L
Drug level			
Digoxin, general range	0.5–1.9 ng/mL	× 1.28	0.6–2.4 nmol/L

SI = US × factor; the SI value is calculated from the US value, except ionized calcium (source in mmol/L). Blood gas values: room air.

Convert US ↔ SI

SI = US × factor; US = SI ÷ factor. mg/dL → mmol/L: factor = 10 ÷ molar mass (g/mol). mEq/L → mmol/L: ÷ ion charge (Na^+, K^+, Cl^-, HCO_3^-: same number). mm Hg → kPa: × 0.1333 (1 mm Hg ≈ 133.322 Pa). HbA1c: mmol/mol = (% − 2.15) × 10.929 (IFCC eq.).

EXAMPLE Three conversions
Glucose 180 mg/dL × 0.0555 = 9.99 ⇒ **10.0 mmol/L**.
Creatinine 1.5 mg/dL × 88.4 = 132.6 ⇒ **133 µmol/L**.
Ionized Ca 1.1 mmol/L ÷ 0.2495 ≈ 4.4088 ⇒ **4.41 mg/dL**.

Unit traps

TRAP Same test, two numbers
Glucose 99 mg/dL = 5.5 mmol/L. Read the unit before you compare a result with a range.

TRAP mEq is not always mmol
Equal only for single-charge ions. Ca^{2+} and Mg^{2+}: 1 mmol = 2 mEq.

TRAP BUN is not urea
US labs report urea **nitrogen** (BUN); SI labs report urea. Urea has 2 N atoms: × 0.357.

TRAP Total vs ionized calcium
Low albumin lowers **total** calcium; ionized calcium (the active fraction) may be normal. Check which one the order and the report give.

TRAP Hemolyzed sample
Red-cell breakdown in the tube releases potassium: a falsely high K^+. Unexpected high K^+: ask whether the sample was hemolyzed and a redraw is needed.

TRAP Hematocrit and SaO_2 in SI
Written as fractions: 37% = 0.37 L/L; 95% = 0.95.

Nursing cues

CLINICAL NOTE INR and bleeding
On warfarin, the usual INR target is 2.0–3.0. Above 3.0: higher bleeding risk. Report per protocol.

CLINICAL NOTE Potassium and the heart
K^+ outside the interval, high or low, can disturb cardiac rhythm: report promptly; ECG monitoring per protocol.

CLINICAL NOTE Digoxin level
The interval shown is a general one; the right level depends on the patient and the indication. Follow the prescriber's target.

Sources: ABIM Laboratory Test Reference Ranges (January 2026); MedlinePlus (INR, digoxin). Factors calculated from IUPAC atomic weights and SI definitions.

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